

File Access

1 Basic Idea

Definition

File access describes the method used to read or write data in a file, especially whether data must be processed in order or can be reached directly.

- Early operating systems mainly supported sequential access.
- Sequential access matched magnetic tape storage.
- Disk storage made random access practical.
- Random access allows programs to jump directly to needed bytes or records.
- Modern systems commonly support both sequential-style reading and explicit repositioning.

2 Sequential Access

Sequential File Access

Sequential access means reading bytes or records in order, starting from the beginning of the file.

- A process reads data in order.
- It cannot skip directly to an arbitrary position.
- It cannot read records out of order.
- The file can usually be rewound.
- After rewinding, the same file can be read again from the beginning.
- This model was convenient for magnetic tape.

Tape Connection

Sequential access fits magnetic tape because tape naturally moves past the read head in order.

3 Random Access

Random-Access File

A **random-access file** is a file whose bytes or records can be read in any order.

- Random access became practical when disks were used for file storage.
- A program can jump to a needed position without reading everything before it.
- Bytes or records can be accessed out of order.
- Some systems also support access by key rather than by numeric position.
- Random access is required by many applications.

4 Why Random Access Matters

Reservation Example

An airline reservation system must access the record for a particular flight directly.

- A customer may ask about one specific flight.
- The reservation program should not read thousands of unrelated flight records first.
- It should jump directly to the needed record.
- This is why database-like applications require random access.

5 Specifying the Start Position

Method	Meaning
Position in every read	Each read operation explicitly gives the file position where reading should begin.
seek operation	A separate operation sets the current file position; later reads continue from that position.

UNIX and Windows

UNIX and Windows use the **seek** model: first set the current position, then read sequentially from there.

6 Seek-Based Access

- The file has a current position.
- A **seek** operation changes that current position.
- After **seek**, the next read starts from the new position.
- Reads can then proceed sequentially from that point.
- This combines random positioning with ordinary sequential reads.

Seek-Then-Read Pattern

```
open(file);  
seek(file, desired_position);  
read(file, buffer, nbytes);
```

7 Sequential vs. Random Access

Feature	Sequential access	Random access
Order	Data are read in order.	Data can be read in any order.
Skipping	Direct skipping is not supported.	Program can jump to a desired location.
Natural medium	Magnetic tape.	Disk and SSD storage.
Typical use	Full file scans, logs, streams.	Databases, indexes, reservation systems.
Current position	Moves forward as data are read.	Can be reset explicitly using position or seek .

8 Key Contrast

- Sequential access is simple and natural for ordered streams.
- Random access is necessary when a program needs one specific record quickly.
- Disks made random access efficient enough for normal file systems.
- The **seek** operation lets a program reposition within a file.
- After seeking, reading can continue sequentially from the new point.

Final Point

Sequential access reads files from beginning to end; random access allows direct movement to the needed byte or record.